

Title

Multimodal Neuroimaging Integration Does Not Enhance Motor Interface Performance in Healthy Learners

Abstract

The combination of scalp electroencephalography and near-infrared spectroscopy was hypothesized to improve brain-machine interface control fidelity compared to electroencephalography alone. Sixteen healthy participants completed five standardized motor control training sessions using either single-source (electroencephalography-only) or dual-source (electroencephalography plus near-infrared spectroscopy) signal interpretation systems. Classification recall measured in real-time during three-class motor tasks averaged 58.2% in the electroencephalography-only condition and 54.1% in the multimodal condition ($p = 0.34$). Agency perception, quantified on standardized scales, approached 73% of maximum in both groups ($p = 0.61$). Computational optimization identified fronto-central electrode placement as most informative across both modalities, while near-infrared sensor positioning required individual-level calibration. Post-hoc simulation analysis indicated comparable predictive utility between modalities. These findings suggest that within standard classification and fusion methodologies, additional neuroimaging channels do not confer performance advantage during acute training periods.